Manish Garinapally

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.59
Narayana junior college	Class XII (TSBIE)	2024	96.10%
Akshara International School	Class X (CBSE)	2022	95.20%

SKILLS

- **Programming Languages:** C, C++, Python, Verilog
- **Simulation & Modeling:** MATLAB/Simulink, Ansys HFSS
- **Circuit Simulation Tools:** LTspice, PSpice, Multisim, Proteus
- **CAD/EDA Software:** Altium Designer, SolidWorks, AutoCAD, Fusion 360, KiCad
- **VLSI/Digital Design Tools:** Cadence, Quartus

Certifications:

Altium Designer Certified (Basic)

Electrical Safety Officer (3E) (up to 3kV)

RELEVANT COURSES

Device physics: semiconductor devices, numerical device modeling, CMOS Fabrication,

Circuit Analysis: Analog Electronics, Digital circuits, signals and systems , Matrix Theory

System Design: System design Hardware Project, PCB design for electronic systems

PROJECTS

BMS Development -- Standalone Wireless Architecture *(Coursework Project)*

- Designed a 18S battery management system using **ADBMS1818 + ESP32** in a standalone configuration **upto 90V**
- Implemented **wireless communication** and **OTA (Over-the-Air)** firmware update capability
- Developed duty-cycle-based cell balancing at **0.5A**, tested and validated on hardware
- Incorporated a high-efficiency(**>90%**) **multistage power management** architecture to independently supply sensing and processing sub-systems

BMS Development -- Multi-Node ASIL-Compliant Daisy-Chain System *(Pravah Racing, Aeolus Racing IIT Hyderabad)*

- Designed a dual-fault-loop BMS architecture using **NXP MC33774 + MC33665 + S32K3**
- Built the system to meet **ASIL** functional safety requirements for automotive-grade reliability with an error margin **< 5%(SoX)**
- Implemented a **multi-nodal daisy-chain** configuration with redundant fault-detection loops for the team's EV battery pack
- Hardware implementation and validation in process

GaN FET-Based Current Source Inverter (CSI) *(Research Project)*

- Designed a CSI using **2EDB8259** gate drivers coupled with **BRB110R2 GaN FETs**
- Achieved **500 kHz** switching speeds through state-of-the-art gate control techniques
- Enabled significant **inductor miniaturization**, reducing passive component footprint in the power conversion stage

Traffic Light Controller (Verilog)

- Designed a traffic light controller in Verilog using a Finite State Machine (FSM) for traffic signal sequencing and

timing control.
- Implemented and verified the logic over multiple iterations of FSMs, optimized to reduce compilation complexity

Three-Way Power Converter -- Multi-Rail LiPo-to-24V/12V/5V System

- Designed a **three-way power converter** delivering **~150W concurrently** across three regulated rails: **24V, 12V, and 5V**, from a **test LiPo battery** source
- Achieved system efficiency **greater than 95%** across all output rails
- Integrated Active power limiting **components** for protection
- Implemented a **Power Good (PG)** output signal per rail for real-time fault detection and immediate system status monitoring

POSITIONS OF RESPONSIBILITY

Core at Aero club for the 2025 -26 Tenure

- developed simulation models for flight control
- Handled Orientations and events

Core at Torque For the 2025-26 Tenure

- Developed and simulated a go-kart chassis, optimizing safety factor and **weight (<15kg)**
- Handled event planning and Execution across multiple revised timelines

TEAM EXPERIENCE

Electrical Head, Pravah Racing (EV) -- Aeolus Racing, IIT Hyderabad *(Nov 2024 - Present)*

- Head the electrical sub-team, leading the design of all Electrical **systems** for the team's 1st electric vehicle
- Designed a **dual-fault-loop BMS** architecture using **NXP MC33774 + MC33665 + S32K3**, built to meet **ASIL** functional safety requirements with error on **SoX < 5%**
- Own fabrication management for LV hardware, achieving a **30,000 cost reduction** through vendor and process optimization
- Apply **DFMA (Design for Manufacturing and Assembly)** principles to coordinate part sequencing and assembly progress across builds
- Coordinate cross-functionally between design and fabrication teams to ensure LV systems meet performance, safety, and reliability standards
- Drive end-to-end ownership of LV systems -- from architecture design through fabrication to on-vehicle integration
- currently Interagting self designed 400V accumulator on the vehicle